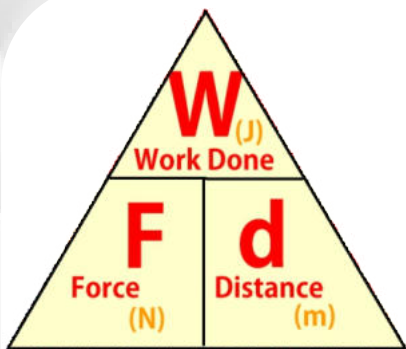


UNIT 08

ENERGY SOURCES AND TRANSFER OF ENERGY



WORK

"The amount of work is the product of force and displacement moved in the direction of force"



UNITS OF WORK

The S.I unit of work is Joule.

$$1 \text{ Joule} = 1 \text{ Nm}$$

Suppose a constant force "**F**" acts on a body and motion takes place in a straight line in the direction of force then work done is equal to the product of magnitude of force "**F**" and the distance "**d**" through which the body moves;

$$W = Fd \cos \phi$$

In that case, we define the work by the force as the product of the component of the force along the line of motion and the distance "**d**"; the body moves along that line, i.e.

Suppose a constant force "**F**" acts on a body

Work = (a component of force) · (distance)

$$W = (F \cos \phi) \cdot d$$

If ϕ between force and displacement is 0, Then $\cos(0) = 1$ and work becomes

$$W = Fd$$

ENERGY

"Energy is a property of matter that allows it to do work. Energy can take many forms, such as heat, light, sound, electrical, chemical, potential and kinetic etc."

UNITS OF ENERGY

The S.I unit of energy is Joule.

KINETIC ENERGY

"Energy possessed by an object due to its motion is called kinetic energy."

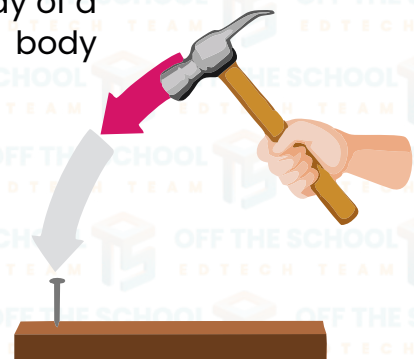
It is also defined as "The work required to accelerate a body of a given mass from rest to its stated velocity". A moving body maintains its kinetic energy unless its speed changes.

UNITS OF KINETIC ENERGY

The S.I unit of kinetic energy is joule (J).

MATHEMATICALLY

$$K.E = \frac{1}{2} m V^2$$



KINETIC ENERGY DEPENDS UPON

The mass m of the object, Greater the mass of the body, greater would be its K.E.

The speed v of the object, Greater the speed of the body, greater would be its K.E.

EXAMPLE

1. Sailing boat.
2. Moving air.
3. Driving a car.

POTENTIAL ENERGY

“The energy that a body possesses by virtue of its position, shape or state of a system.”

There are different types of potential energy. Like gravitational potential energy, elastic potential energy, and chemical potential energy.

UNIT

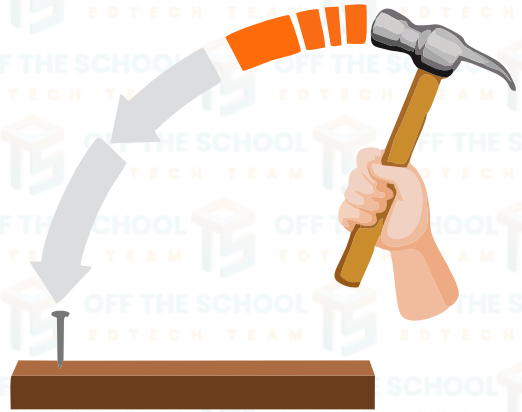
The S.I unit of potential energy is joule (J).

- A body raised to a height “ h ” above the ground has gravitational potential energy.
- A stretched spring has elastic potential energy due to its stretched position (condition).
- The energy stored in the plants that we eat is chemical potential energy

It's also defined as the work done stored in a body in lifting it to height “ h ”. The potential energy changes only when its position relative to the ground changes; otherwise it remains same.

MATHEMATICALLY

$$P.E = mgh$$

**DERIVATION OF GRAVITATIONAL POTENTIAL ENERGY $P.E = MGH$**

To derive the expression for the gravitational potential energy, let us consider an object of mass “ m ” which is raised up through height “ h ” from the ground. The work done in lifting it to height “ h ” is stored in it as its gravitational potential energy “ $P.E$ ”.

$$P.E = \text{Work done}$$

$$P.E = W$$

$$P.E = F.d \quad \text{----- (i)}$$

As,

$$F = mg$$

$$d = h$$

Equation (i) becomes,

$$P.E = (mg).h$$

$$P.E = mgh$$

CONSERVATION OF ENERGY

Energy can neither be created nor destroyed, but it can be converted from one form to another. This is called the law of conservation of energy.

CONVERSION OF ENERGY FROM ONE FORM TO ANOTHER

FOSSIL FUEL ENERGY

Fossil fuel energy is produced from ancient plants and animals that have undergone a transformation into crude oil, coal, natural gas, or heavy oils due to heat and pressure in the Earth's crust over many millions of years. These fossil fuels store a significant amount of chemical energy. This stored energy is then converted into heat and light energy through oxidation, also known as burning. When a fossil fuel such as charcoal is burned, it releases heat and light energy.

**HYDROELECTRIC ENERGY**

Hydroelectricity is a type of electricity that is generated through the use of hydropower. This involves harnessing the gravitational force of flowing or falling water to generate power. One of the most widely used methods of producing hydroelectricity is through the use of a dam on a river. This involves storing water in a reservoir and then releasing it to flow through a turbine, which spins and drives a generator to produce electricity.

**SOLAR ENERGY**

Solar energy refers to the energy that is radiated from the sun and is abundant in Pakistan. There are various devices that can harness this solar energy and convert it into either electrical energy or heat energy. Examples include photovoltaic solar panels and solar cells, which convert the sun's rays into electricity for various purposes. Solar heaters are also used to convert solar energy into heat energy, which can be used to heat water tanks and indoor spaces.

**GEOTHERMAL ENERGY**

Geothermal energy is stored in the Earth as natural heat, which originates from the hot molten part called magma deep in the Earth. When water is in close proximity to the magma, it becomes steam due to the high temperature and this thermal energy is conducted to the Earth's surface. A geothermal power plant harnesses this energy to generate electricity through the use of an electrical generator. Geothermal wells can be created by drilling deep near hot rocks at various locations where magma is close to the surface. The hot rocks heat the water that is pumped into the well, causing it to change into steam. This steam can be used either for heating or to power an electrical generator.

**WIND ENERGY**

The energy obtained by the wind is called wind energy. It is generated by windmills. A windmill consists of a turbine that rotates due to wind. Kinetic energy is produced due to the motion of the turbine.



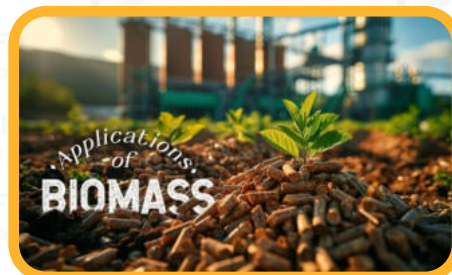
Wind turbines convert this kinetic energy into mechanical power. A generator converts that mechanical power into electricity.

APPLICATIONS

It is being used as the source of energy for sailing ships in oceans. It is being used by windmills to pump water. It is being used by windmills to grind grain. It is used to turn wind turbines to produce electricity.

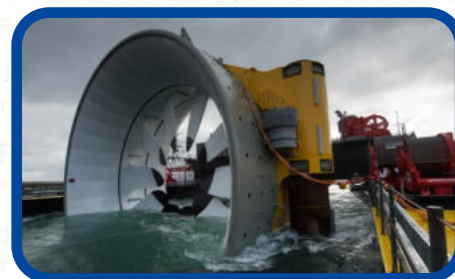
BIOMASS ENERGY

Biomass is the organic material that comes from plants and animals. Biomass consists of stored energy from the Sun, garbage, wastes, sugarcane, etc. Solid biomass, such as wood, organic material, and garbage, can be burned directly to produce heat. Biomass can also convert into a gas called biogas and into liquid biofuels such as ethanol and biodiesel.



TIDAL ENERGY

It is a form of hydropower that converts the energy obtained from tides into a useful form of power mainly electricity as the Earth uses the gravitational forces of both the moon and the sun every day to move vast quantities of water around the oceans and seas producing tides and in this way energy is produced called tidal energy.



GEOTHERMAL ENERGY

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Renewable Energy Source

- ① Renewable resources are replenished naturally and at a rate that is equal to or faster than their rate of consumption.
- ② Renewable energy sources are clean and produce little to no pollution.

Non-Renewable Energy Source

Non-renewable resources are finite and once they are used, they cannot be replenished.

Non-renewable energy sources are often associated with pollution and negative environmental impact.

3 Renewable energy sources, such as wind, solar, and hydropower, are widely available and can be generated in many parts of the world.

Non-renewable energy sources, such as coal, oil, and natural gas, are often limited to specific geographic locations and can be subject to supply disruptions.

EFFICIENCY AND POWER

A system in which some energy 'E₁' is supplied to it as input and the system returns back some energy 'E₂' as output has some efficiency. This efficiency is defined as:

"The ratio of output to the input is called Efficiency."

UNIT

It is a ratio of two energies therefore it has no unit. No machine is 100% efficient because some energy is always wasted in the form of heat, sound or light etc.

$$\text{Efficiency} = \frac{\text{Energy as output}(E_1)}{\text{Energy as input}(E_2)}$$

$$\text{Efficiency} = \frac{E_2}{E_1} \times 100$$

$$\text{Efficiency} = \frac{\text{Output}}{\text{Input}} \times 100$$

POWER

"The rate of doing work is called Power"

Or

"The amount of energy transferred per unit time is called Power."

EXPLANATION

When you run up and cover distance in 5 seconds or take a slow walk up the same distance in 20 seconds. You are doing the same amount of work; however, you are doing it at a different rate. When you run up, you are working much faster and you have a higher power than when you walk up. This quantity tells us the rate of doing work

$$\text{Power} = \frac{\text{Work done}}{\text{time taken}}$$

$$P = \frac{W}{t}$$

Since work and time are scalar quantities. Therefore, power is also a scalar quantity.

UNIT OF POWER

The S.I unit of power is $\frac{\text{joules}}{\text{second}} = \text{Watt}$

LARGER UNITS OF POWER ARE

Kilowatt (kW) or 1kW = 1000 Watt

1 MW = 1000000 Watt

1 hp = 746 Watts